ROBOTICS 4/M

TOPIC 6: DIFFERENTIAL KINEMATICS AND VELOCITIES

6.1 INTRODUCTION

Differential kinematics are small movements of any mechanism (e.g. robot manipulators) in short periods of time. These motions can be used to derive the velocity relationships between different parts of a robot, in a similar way to the kinematic relationships studied previously define the displacement relationships of a robot [1,2]. These differential kinematics describe the velocity related motions of frames relative to fixed frames.

6.2 DIFFERENTIAL MOTIONS OF A FRAME

We define the differential kinematics of a robot by the combined differential motions of the joints that comprise the robot and its end effector. To do this we must consider the differential motions of the frames associated with the respective joints of the robot. As with the kinematic relationships, the differential motions of frames are defined in terms of the different types of relationships [1]:

- Differential Translation
- Differential Rotation
- Differential Transformations (combinations)

These are described below

DIFFERENTIAL TRANSLATION: The differential translation is a translation of a frame by a relatively small amount along all three axes [1,2]. This is represented in the following way:

$$Trans(dx, dy, dz) = \begin{bmatrix} 1 & 0 & 0 & dx \\ 0 & 1 & 0 & dy \\ 0 & 0 & 1 & dz \\ 0 & 0 & 0 & 1 \end{bmatrix} \tag{6.1}$$

DIFFERENTIAL ROTATION: The differential rotations are defined in terms of small differential rotations about the *x*, *y* and *z* axes [1,2]. The differential motions are represented by small angles δx, δy, δz. The transformation matrices for kinematics is used again. However, since the angles are small the following approximations are applied (only in radians):

$$\sin \delta x \approx \delta x$$
$$\cos \delta x \approx 1$$

It follows that the resulting rotation transformation matrices for rotations about the *x*, *y* and *z* axes are:

$$Rot(x, \delta x) = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & -\delta x & 0 \\ 0 & \delta x & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} \tag{6.2}$$

$$Rot(y, \delta y) = \begin{bmatrix} 1 & 0 & \delta y & 0 \\ 0 & 1 & 0 & 0 \\ -\delta y & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} \tag{6.3}$$

$$Rot(z, \delta z) = \begin{bmatrix} 1 & -\delta z & 0 & 0 \\ \delta z & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} \tag{6.4}$$

If we consider multiple differential rotations, e.g. rotation about *x*-axis followed by rotation about *y*-axis, gives different results depending on the order of rotation e.g.

$$Rot(x, \delta x)Rot(y, \delta y) = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & -\delta x & 0 \\ 0 & \delta x & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} \cdot \begin{bmatrix} 1 & 0 & \delta y & 0 \\ 0 & 1 & 0 & 0 \\ -\delta y & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} = \begin{bmatrix} 1 & 0 & \delta y & 0 \\ \delta x \delta y & 1 & -\delta x & 0 \\ -\delta y & \delta x & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} \tag{6.5}$$

$$Rot(y,\delta y)Rot(x,\delta x) = \begin{bmatrix} 1 & 0 & \delta y & 0 \\ 0 & 1 & 0 & 0 \\ -\delta y & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} \cdot \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & -\delta x & 0 \\ 0 & \delta x & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} = \begin{bmatrix} 1 & \delta x\delta y & \delta y & 0 \\ 0 & 1 & -\delta x & 0 \\ -\delta y & \delta x & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} \tag{6.6}$$

The results are quite different. However, since the angles are small the multiplication of multiple angles will be negligibly small. This would mean that the $\delta x\delta y$ term will become zero in both [1]. This would mean that:

$$Rot(x,\delta x)Rot(y,\delta y) = Rot(y,\delta y)Rot(x,\delta x) = \begin{bmatrix} 1 & 0 & \delta y & 0 \\ 0 & 1 & -\delta x & 0 \\ -\delta y & \delta x & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} \tag{6.7}$$

This means that if the small angle multiplication rule is applied the order of multiplication is not important.

DIFFERENTIAL ROTATION ABOUT A GENERAL AXIS: Since the order of rotation is not important for differential rotations, the transformation matrices can be multiplied in any order. If we assume that the differential rotation of $d\theta$ about any general axis $\boldsymbol{k}$ can be represented by differential rotations about the three axes in any order, we get the following relationship [1]:

$$\begin{aligned} Rot(k,d\theta) &= Rot(x,\delta x)Rot(y,\delta y)Rot(z,\delta z) \\ &= \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & -\delta x & 0 \\ 0 & \delta x & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} 1 & 0 & \delta y & 0 \\ 0 & 1 & 0 & 0 \\ -\delta y & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} 1 & -\delta z & 0 & 0 \\ \delta z & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} \\ &= \begin{bmatrix} 1 & -\delta z & \delta y & 0 \\ \delta x\delta y + \delta z & -\delta x\delta y\delta z + 1 & -\delta x & 0 \\ -\delta y + \delta x\delta z & \delta x + \delta y\delta z & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} \end{aligned} \tag{6.8}$$

If we neglect multiple angles, Equation (6.8) can be simplified as follows:

$$Rot(k,d\theta) = Rot(x,\delta x)Rot(y,\delta y)Rot(z,\delta z) = \begin{bmatrix} 1 & -\delta z & \delta y & 0 \\ \delta z & 1 & -\delta x & 0 \\ -\delta y & \delta x & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} \tag{6.9}$$

This gives the total differential transformations about a general axis $\boldsymbol{k}$.

DIFFERENTIAL TRANSFORMATIONS OF A FRAME: The differential transformation of a frame is a combination of differential translations and rotations about the x, y, z- axes [1,2]. If we denote the frame as matrix T and the differential change in the frame as dT, then

$$[T + dT] = [Trans(dx,dy,dz)Rot(k,d\theta)][T] \tag{6.10}$$

By rearranging we get:

$$[dT] = [\Delta][T] = [Trans(dx,dy,dz)Rot(k,d\theta) - I][T] \tag{6.11}$$

Δ is called the *differential operator* and is calculated thus

$$\begin{aligned} \Delta &= Trans(dx,dy,dz)Rot(k,d\theta) - I \\ &= \begin{bmatrix} 1 & 0 & 0 & dx \\ 0 & 1 & 0 & dy \\ 0 & 0 & 1 & dz \\ 0 & 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} 1 & -\delta z & \delta y & 0 \\ \delta z & 1 & -\delta x & 0 \\ -\delta y & \delta x & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} - \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} = \begin{bmatrix} 0 & -\delta z & \delta y & dx \\ \delta z & 0 & -\delta x & dy \\ -\delta y & \delta x & 0 & dz \\ 0 & 0 & 0 & 0 \end{bmatrix} \end{aligned} \tag{6.12}$$

As you may notice that the differential operator is not a transformation matrix or a frame as it is not in the form needed to define either of these. This is simply an operator to define the change in a frame [1].

To define the change in the frame we define the dT matrix in terms of the differential change in the n,o,a frame axes and the position of the frame origin i.e.

$$[dT] = \begin{bmatrix} dn_x & do_x & da_x & dp_x \\ dn_y & do_y & da_y & dp_y \\ dn_z & do_z & da_z & dp_z \\ 0 & 0 & 0 & 0 \end{bmatrix} \tag{6.13}$$

Therefore, from Equation (6.11)

$$\begin{bmatrix} dn_x & do_x & da_x & dp_x \\ dn_y & do_y & da_y & dp_y \\ dn_z & do_z & da_z & dp_z \\ 0 & 0 & 0 & 1 \end{bmatrix} = \begin{bmatrix} 0 & -\delta z & \delta y & dx \\ \delta z & 0 & -\delta x & dy \\ -\delta y & \delta x & 0 & dz \\ 0 & 0 & 0 & 0 \end{bmatrix} \cdot T \tag{6.14}$$

6.3 Differential Changes Between Frames

So far the differential operator has been defined with respect to the fixed (universal) reference frame. It is possible to define another differential operator relative to the current moving frame i.e. ${}^{T}\Delta$. To find these changes to the frame we must postmultiply the frame matrix by ${}^{T}\Delta$. This will give the same result as they both define the changes to the frame. It follows that [1]:

$$[dT] = [\Delta][T] = [T]\left[{}^{T}\Delta\right] \tag{6.15}$$

We can obtain this moving frame differential operator by premultiplying by the inverse of the frame matrix i.e.

$$\begin{aligned} [T]^{-1}[\Delta][T] &= [T]^{-1}[T]\left[{}^{T}\Delta\right] \\ \left[{}^{T}\Delta\right] &= [T]^{-1}[\Delta][T] \end{aligned} \tag{6.16}$$

Using the previous information on inverse transformation matrices and the definition of the differential operator in (6.12) we get:

$$\begin{aligned} \left[{}^{T}\Delta\right] &= [T]^{-1}[\Delta][T] \\ &= \begin{bmatrix} n_x & n_y & n_z & -p\cdot n \\ o_x & o_y & o_z & -p\cdot o \\ a_x & a_y & a_z & -p\cdot a \\ 0 & 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} 0 & -\delta z & \delta y & dx \\ \delta z & 0 & -\delta x & dy \\ -\delta y & \delta x & 0 & dz \\ 0 & 0 & 0 & 0 \end{bmatrix} \begin{bmatrix} n_x & o_x & a_x & p_x \\ n_y & o_y & a_y & p_y \\ n_z & o_z & a_z & p_z \\ 0 & 0 & 0 & 1 \end{bmatrix} \\ &= \begin{bmatrix} 0 & -{}^{T}\delta z & {}^{T}\delta y & {}^{T}dx \\ {}^{T}\delta z & 0 & -{}^{T}\delta x & {}^{T}dy \\ -{}^{T}\delta y & {}^{T}\delta x & 0 & {}^{T}dz \\ 0 & 0 & 0 & 0 \end{bmatrix} \end{aligned} \tag{6.16}$$

Here the entries for the moving frame differential operator are [1]:

$$\begin{aligned} {}^{T}\delta x &= \bar{\delta}\cdot\bar{n} \\ {}^{T}\delta y &= \bar{\delta}\cdot\bar{o} \\ {}^{T}\delta z &= \bar{\delta}\cdot\bar{a} \\ {}^{T}dx &= \bar{n}\cdot\left[\left(\bar{\delta}\times\bar{p}\right)+\bar{d}\right] \\ {}^{T}dy &= \bar{o}\cdot\left[\left(\bar{\delta}\times\bar{p}\right)+\bar{d}\right] \\ {}^{T}dz &= \bar{a}\cdot\left[\left(\bar{\delta}\times\bar{p}\right)+\bar{d}\right] \end{aligned} \tag{6.17}$$

A complete derivation of these relationships is given in [3]

References

[1] Niku, S.B., *Introduction to Robotics*, Pearson Education, 2001.
[2] Sciavicco, L. & Siciliano, B., *Modelling and Control of Robot Manipulators*, Springer, 2001.
[3] Paul, R.P., *Robot Manipulators, Mathematics, Programming and Control*, The MIT Press, 1981.